Pentafecta v3

Advanced Multi-Stage Traffic Demand Forecasting System

A production-grade, stacking-ensemble framework engineered for high-precision spatial-temporal traffic prediction — combining gradient boosting, probabilistic modeling, and harmonic time-series decomposition into a unified forecasting pipeline.

LEAD DATA SCIENCE | URBAN MOBILITY ANALYTICS | ML ENGINEERING

Executive Summary

Why Pentafecta v3 Exists

Traffic demand forecasting is a high-stakes spatial-temporal problem. Pentafecta v3 delivers **sub-hourly demand predictions** across urban networks by fusing ensemble learning, geospatial clustering, and harmonic time-series decomposition into a single, production-ready pipeline.

🎯 Precision

Log-transformed targets and 5-fold OOF predictions minimize RMSE on skewed demand distributions.

🧩 Ensemble Power

Four heterogeneous base learners — LightGBM, CatBoost, XGBoost, NGBoost — blended via RidgeCV meta-learner.

📍 Spatial Intelligence

Geohash-level statistics and K-Means behavioral clustering (15 clusters) encode road-network topology.

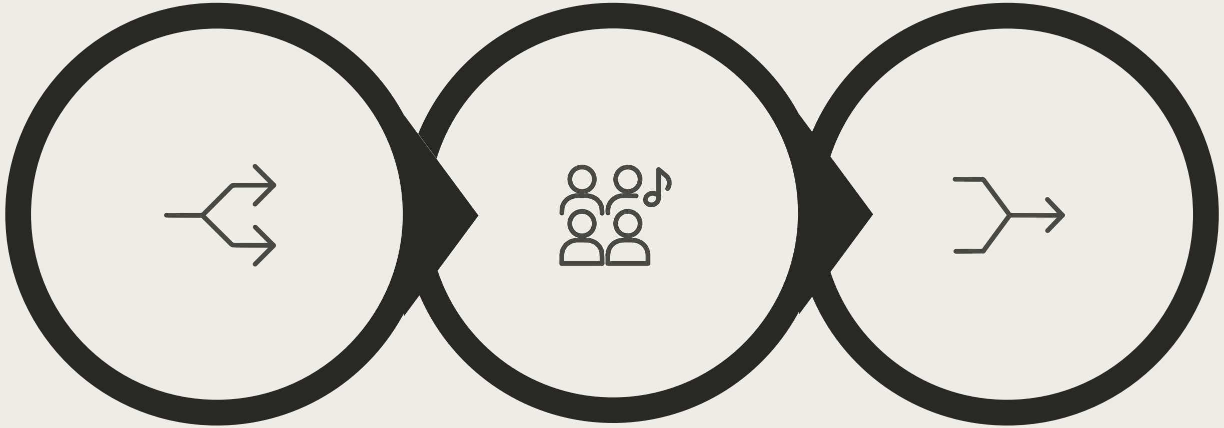
⏱️ Temporal Depth

Harmonic sine/cosine transforms capture 24h, 12h, and 6h cycles. Prophet trend signals serve as meta-features.

Model Architecture

Stacking Ensemble Strategy

Pentafecta v3 uses a **two-level stacking ensemble**. Level-0 generates Out-of-Fold (OOF) predictions from four base learners via 5-fold CV. Level-1 trains a RidgeCV meta-learner on OOF outputs to produce the final forecast — controlling overfitting while maximizing model diversity.



5-Fold CV

Base Learners

RidgeCV Meta

OOF predictions ensure the meta-learner trains on unbiased estimates, preserving generalization on unseen data.

Base Learners

Four Heterogeneous Models, One Unified Forecast

LightGBM

Leaf-wise tree growth with histogram-based splitting. Excels on high-cardinality categorical features and large-scale tabular data with minimal memory footprint.

CatBoost

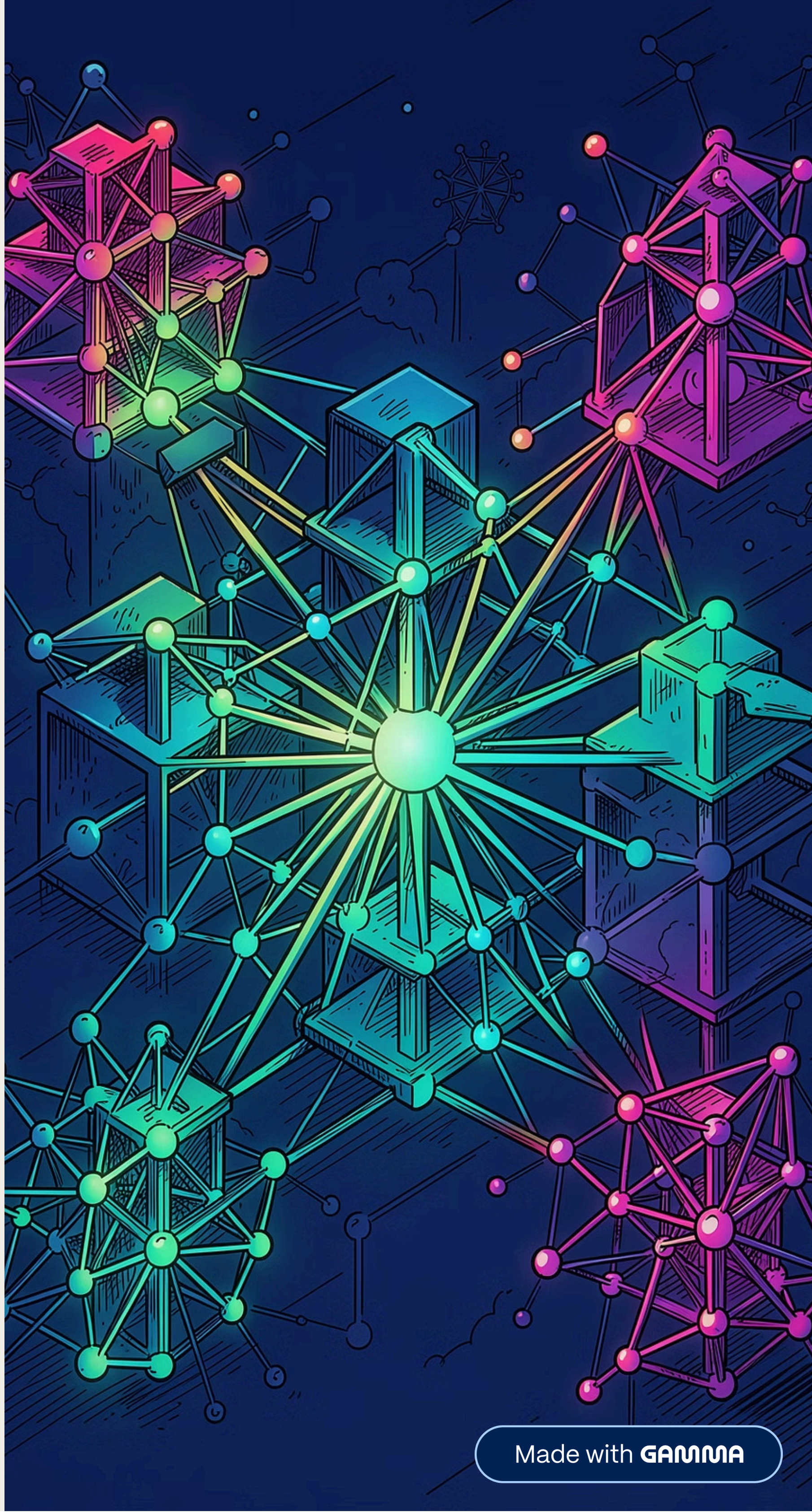
Native categorical handling via ordered target statistics. Ordered boosting reduces prediction shift, providing robust performance on location-encoded features.

XGBoost

Second-order gradient optimization with regularization. Strong baseline for tabular regression with proven convergence on structured traffic features.

NGBoost

Natural Gradient Boosting for probabilistic forecasts. Outputs full predictive distributions (mean + variance), capturing demand uncertainty for downstream planning.



Meta-Learner

RidgeCV: The Final Blending Layer

The Level-1 meta-learner is a **Ridge Regression with built-in Cross-Validation (RidgeCV)**. It takes the OOF prediction matrix from all four base learners and learns optimal blending weights that minimize squared error.

Ridge regularization prevents overfitting when base learner outputs are correlated. The internal CV automatically selects the optimal α penalty, eliminating manual hyperparameter tuning.

> RidgeCV is ideal here: base learner outputs are continuous, roughly Gaussian (post log-transform), and moderately correlated — exactly the regime where linear blending with L2 regularization outperforms non-linear meta-learners.

Blending Formula

Final prediction is a regularized linear combination:

$$\hat{y}_{final} = \sum_{i=1}^{4} w_i \cdot \hat{y}_{OOF}^{(i)}$$

Weights w_i are learned via RidgeCV with automatic α selection across a logarithmic grid.

Feature Engineering · 36 Features

Spatial Features: Encoding Where Traffic Happens

Geohash-Level Statistics

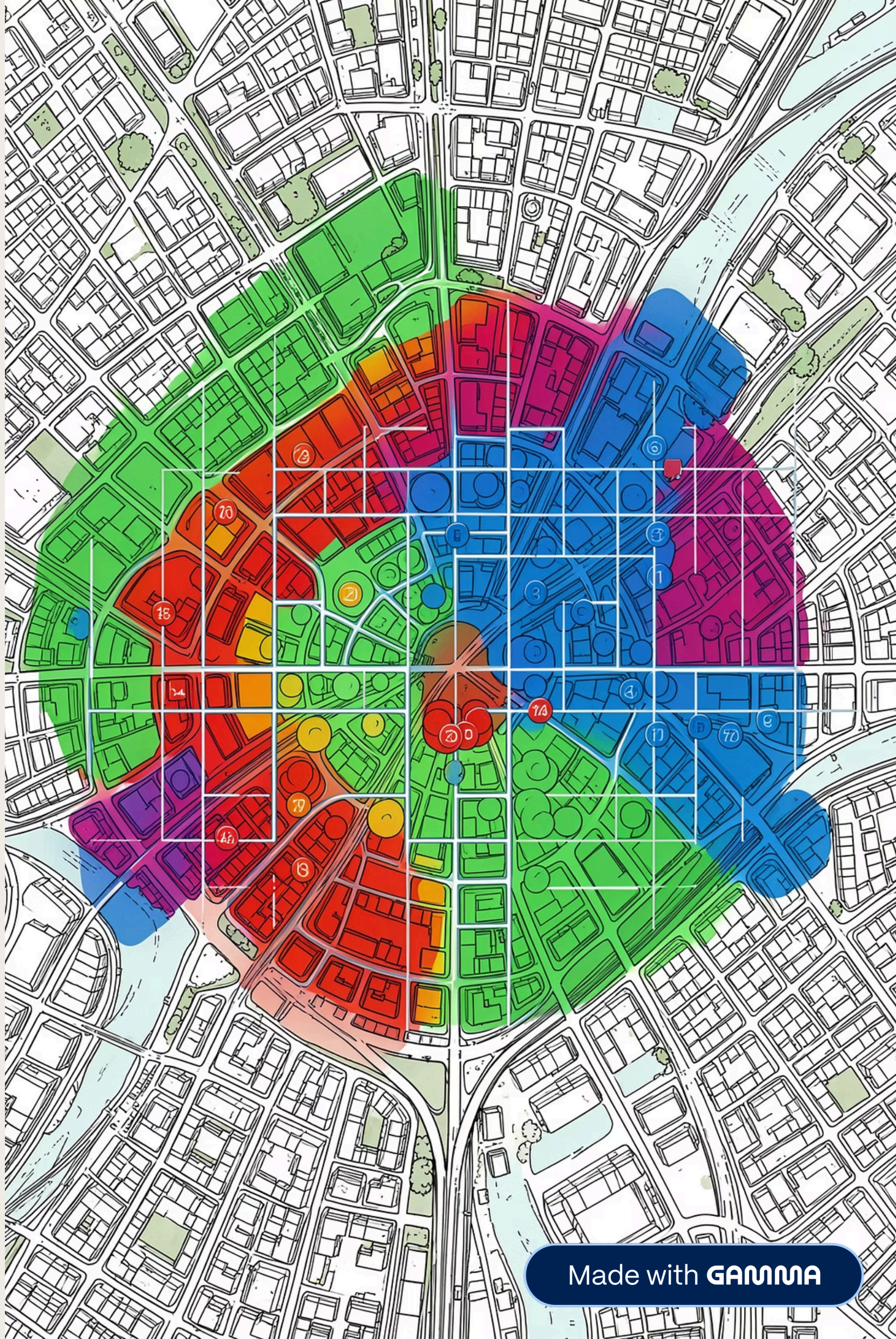
Road segments are encoded into **geohash prefixes** (precision 6–7), then aggregated into statistical features: mean demand, standard deviation, min/max, and demand skewness per geohash cell. These capture local traffic baselines and variability patterns.

- Geohash mean & median demand
- Geohash demand std dev & range
- Neighbor geohash differential features

Behavioral Clustering (K-Means, k=15)

A **K-Means clustering model** (k=15) is fitted on historical demand profiles per segment. Each segment receives a cluster label and distance-to-centroid features, encoding behavioral archetypes:

- Cluster assignment (categorical → target encoded)
- Distance to each of 15 centroids
- Cluster-level demand statistics as cross-features



Feature Engineering · 36 Features

Temporal Features: Capturing When Traffic Peaks

Traffic demand is inherently periodic. Pentafecta v3 uses **harmonic transformations** — sine and cosine pairs — to encode cyclical time patterns without discontinuities at period boundaries.

24-Hour Cycle

sin(2π·hour/24) and **cos(2π·hour/24)**
Captures daily rush-hour patterns: morning peak (7–9am) and evening peak (5–7pm).

12-Hour Cycle

sin(2π·hour/12) and **cos(2π·hour/12)**
Encodes AM/PM symmetry, distinguishing morning vs. afternoon demand asymmetries.

6-Hour Cycle

sin(2π·hour/6) and **cos(2π·hour/6)**
Captures sub-daily micro-patterns: late-night lulls, midday plateaus, and transitional periods.

Day-of-Week

One-hot + harmonic encoding for weekday vs. weekend behavioral shifts. Critical for modeling Friday evening surges and Monday morning ramps.

Feature Engineering · 36 Features

Interaction Features: High-Order Crosses

Density Proxies

Interactions between geohash demand levels and temporal windows create **density proxy features** — estimating localized congestion intensity without explicit sensor data.

- Geohash mean × hour-of-day harmonic
- Cluster centroid distance × day-of-week
- Segment demand ratio × temporal cycle

Environmental-Spatial Interactions

Crosses between spatial cluster membership and temporal signals capture **context-dependent demand shifts** — e.g., how a commercial district cluster behaves differently at 8am vs. 8pm.

- Cluster × 24h harmonic cross-terms
- Geohash rank × weekend indicator
- Demand volatility × temporal phase

All 36 features undergo variance inflation factor (VIF) screening to remove multicollinear interactions before final model training.

Time-Series Integration

Prophet Signals as Meta-Features

Beyond harmonic features, Pentafecta v3 integrates **Facebook Prophet** as an auxiliary time-series model. Prophet decomposes each segment's historical demand into trend, seasonality, and holiday components — then its forecasts are ingested as **meta-features** into the stacking ensemble.

Long-Term Trend Signal

Prophet's piecewise linear trend captures structural shifts in demand (e.g., post-pandemic recovery, new transit infrastructure) that tree-based models may underweight.

Multi-Seasonality Decomposition

Prophet models weekly and yearly seasonality explicitly, complementing harmonic features with additive seasonal components validated against holdout periods.

Holiday & Event Effects

Custom holiday regressors in Prophet capture demand anomalies around public holidays, major events, and school schedules — encoded as residual meta-features for the ensemble.

Training & Deployment

From Log-Transform to Production Pipeline

Training Methodology

Traffic demand is right-skewed with heteroscedastic variance. Pentafecta v3 applies **log1p transformation** to the target before training, stabilizing variance and normalizing the residual distribution.

- **log1p(y)** target transform → train on log-scale
- **exp(pred) − 1** inverse transform for final output
- **5-Fold CV** for OOF prediction generation
- Stratified folds by geohash to preserve spatial distribution

Performance Optimization

RMSE is the primary optimization metric. Hyperparameters for each base learner are tuned via **Bayesian optimization (Optuna)** with early stopping to prevent overfitting.

Production Pipeline

The full system is consolidated into a **single Python pipeline** (scikit-learn Pipeline + custom transformers):

01

Raw Input

Segment IDs, timestamps, historical demand

02

Feature Transformer

Spatial, temporal, interaction, Prophet features

03

Ensemble Predict

4 base learners → RidgeCV blend

04

Inverse Transform

exp(pred)−1 → final demand forecast